Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[ ]beamerthemeAalborg
[utf8]inputenc [english]babel [T1]fontenc helvet

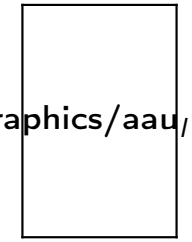
The Aalborg Beamer Theme

v. 1.3.0

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19 Pebruari 2025





Agenda

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Introduction

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- Since there was no Aalborg University (AAU) branded beamer theme, I published the theme after the conference on my website so that other researches and students at AAU could use the theme for their presentations.
- I have received a lot of positive feedback. To my surprise, several people not affiliated with AAU have used the theme despite the heavy AAU branding.
- To make the theme more usable to people not affiliated with AAU, I have decided to create this theme called Aalborg which is very similar to the AAU sidebar theme, but it does not come with the heavy AAU branding.

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License

- The AAU logo is covered by copyright rules. I have used the logo from . As long as you use the theme for making presentations in connection with your work at AAU, you are allowed to use the AAU logo.

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- The rest of the theme is provided under the GNU General Public License v. 3 (GPLv3). This basically means that you can redistribute it and/or modify it under the same license. For more information on the GPL license see

Installation

The theme consists of four files

1. `beamerthemeAalborg.sty`
2. `beamerinnerthemeAalborg.sty`
3. `beameroutertheme Aalborg.sty`
4. `beamercolorthemeAalborg.sty`

The theme can either be installed for local or global use.

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Local Installation

The simplest way of installing the theme is by placing the four theme files in the same folder as your presentation. When you download the theme, the four theme files are located in the `local` folder.

Installation

Global Installation

- If you wish to make the theme globally available, you must put the files in your local latex directory tree. The location of the root of the local directory tree depends on the operating system and the latex distribution. On the following slides, you can read the instructions for some common setups.
- When you download the theme, the four theme files are embedded in a directory structure (in the `global` folder) ready to be copied directly to the root of your local directory tree.
- On the following slides, we refer to this directory structure as `<dirstruct>`. Note that some parts of the directory may already exist if you have installed other packages in your local latex directory tree. If this is the case, you simply merge `<dirstruct>` with your existing setup.

Installation

GNU/Linux

Ubuntu with TeX Live

1. Place the `<dirstruct>` in the root of your local latex directory tree. By default it is
   `~/texmf`
   If the root does not exist, create it. The symbol ~ refers to your home folder, i.e., `/home/<username>`
2. In a terminal run
   `$ texhash ~/texmf`

Windows with MiKTeX

Apparently, MiKTeX does not include a local latex directory tree by default. Therefore, you first have to create it.

1. To do this, create a folder `<somewhere>` named, e.g., `texmf`
2. Add this folder in the Roots tab of the MiKTeX Settings dialog
3. Place the `<dirstruct>` in your newly created local latex directory tree
   `<somewhere>\texmf`
4. Open the MiKTeX Settings dialog and click Refresh FNDB.

Windows with TeX Live

In the advanced TeX Live Installer, you can manually change the default position of the root of the local latex directory tree. However, we assume the default position below.

1. Place the `<dirstruct>` in your local latex directory tree `%USERPROFILE%\texmf`
   If it does not exist, create it. In XP `%USERPROFILE%` is `c:\Document and Settings\<username>`
   by default, and in Vista and above it is by default `c:\Users\<username>`
2. Open the TeX Live Manager dialog and select 'Update filename database' under 'Actions'.

Installation

Mac OS X

Mac OS X with MacTeX

Place the `<dirstruct>` in the root of your local latex directory tree. By default it is
`~/Library/texmf`
If the root does not exist, create it. The symbol ~ refers to your home folder, i.e., `/home/<username>`

Installation

Required Packages

Of course, you have to have the Beamer class installed. In addition, the theme loads two packages

- TikZ[1]
- calc

These packages are very common and should therefore be included in your latex distribution.

[1] By the way, TikZ is an awesome package for creating beautiful graphics. If you do not believe me, then have a look at these or the AAUgraphics/aau/. If you want to create beautiful plots, you should use the pgfplots package which is based on TikZ.

The Presentation Theme

It is very simple to load the presentation theme. Just type
`\usetheme[<options>]{Aalborg}`
which is exactly the same way other beamer presentation themes are loaded. The presentation theme loads the inner, outer and color Aalborg theme files and passes the `<options>` on to these files.

The Inner Theme

You can load the inner theme directly by
`\useinnertheme{Aalborg}`
and it has no options.

The Outer Theme

You can load the outer theme directly by

`\useoutertheme[<options>]{Aalborg}`

Currently, the theme options are

- `hidetitle`: Hide the (short) title in the sidebar
- `hideauthor`: hide the (short) author in the sidebar
- `hideinstitute`: hide the (short) institute in the bottom of the sidebar
- `shownavsym`: show the navigation symbols
- `left` or `right`: position of the sidebar (default is right)
- `width=<length>`: width of the sidebar (default is 2 cm).
- `hideothersubsections`: hide all subsections but the subsections in the current section
- `hideallsubsections`: hide all subsections

The last four options are inherited from the outer sidebar theme.

User Interface

Loading the Theme and Theme Options

The Color Theme

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`\usecolortheme{Aalborg}`

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The Color Element `Aalborg`

The color theme defines a new beamer color element named `Aalborg` whose foreground and background colors are

- fg: light blue ({RGB}{194,193,204})
- bg:

You can use these colors in the standard beamer way by using the command `\usebeamercolor[<fg or bg>]{Aalborg}`. See the beamer manual for instructions.

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You can use these colors in the standard beamer way by using the command `\usebeamercolor[<fg or bg>]{Aalborg}`. See the beamer manual for instructions.
Note that this version of the theme is an official AAU version and is in accordance with the . However, you can easily change it (including the colour of the logo) by following the steps in `beamercolorthemeAalborg.sty`.



Compiling Your Presentation With the Aalborg Theme

Unlike most other beamer themes, this theme must be compiled at least three times to make everything look right. For most other themes, you do not have to compile your presentation more than two times. For the Aalborg theme, the third compilation is necessary to determine the position of the circle with the current frame number and the length of the "progress bar".

User Interface

Modifying the Theme

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User Interface

Modifying the Theme

- The default configuration of fonts, colors, and layout complies with the AAU design guidelines and is the official version of the theme.
- However, you can modify specific elements of the theme through the templates provided by the beamer class. Please refer to the beamer user manual for instructions.
- For example, on this slide the following commands have been used
  - Change the bar and sidebar colors:
    ```
    \setbeamercolor{Aalborg}{fg=gray!50,bg=gray}
    \setbeamercolor{sidebar}{bg=red!20}
    ```
  - Change the color of the structural elements:
    ```
    \setbeamercolor{structure}{fg=red}
    ```
  - Change the frame title text color:
    ```
    \setbeamercolor{frametitle}{use=structure,
    fg=structure.fg}
    ```
  - Change the background color of the text
    ```
    \setbeamercolor{normal text}{bg=gray!20}
    ```

The AAU Waves Background Image

- In this documentation, the title page frame and the last frame have the AAU waves as the background image. The AAU waves background image can be added to any single frame by wrapping a frame in the following way

```
{\aauwavesbg
\begin{frame}[<options>]{Frame Title}{Frame
Subtitle}
...
\end{frame}}
```

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```

- Ideally, I would like to create a new frame option called `aauwavesbg` which can enable the AAU waves background. However, I have not been able to figure out how such an option can be added. If you know how this can be done, please contact me.

Widescreen Support

Newer projectors and almost any modern TV support a widescreen format such as 16:10 or 16:9. Beamer (>= v. 3.10) supports various aspect ratios of the slides. According to section 8.3 on page 77 of the Beamer user guide v. 3.10, you can write
`\documentclass[aspectratio=1610]{beamer}`
to get slides with an aspect ratio of 16:10. You can also use 169, 149, 54, 43 (default), and 32 to get other aspect ratios.

Feedback

Known Problems

Overlapping footnote You might have problems with a too wide footnote text width. This is problem with older versions of Beamer, and it can be resolved by updating Beamer to a newer version. You can read more about it in .

Feedback

Bugs, Comments and Suggestions

- There are probably still some bugs in the theme. If you should find one, then please let me know. No bug is too small!

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- There are probably still some bugs in the theme. If you should find one, then please let me know. No bug is too small!
- Also, please contact me, if you have some exciting new ideas or just some simple usability improvements.

Feedback

Contact Information

In case you have any comments, suggestions or have found a bug, please do not hesitate to contact me. You can find my contact details below.

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jkn@es.aau.dk

Niels Jernes Vej 12, A6-309
9220 Aalborg Ø

Thank you for using this theme!